‘Creative Thinking’ for AI and IoT Innovations

Introduction: Creativity, thinking and AI

Parag Kulkarni, PhD, DSc

What is creativity?

- Is it standardized?
- Is it very personal?
- Can it be separate for humans and machines?
- Novelty and creativity - difference?
- Creative thinking in AI age
- Creative thinking for AI innovations – AI innovations for creative thinking
- AI for creativity – Creativity for AI

What this course is about?

- Basics of AI and IOT
- Machine creativity – debate and possibility
- Machine's ability to help humans while performing something that fall under creativity discipline
- Co-working between humans and machines for creativity
- Algorithms those could aid creativity
- History of Creativity and Technology
- AI augmented creativity
- AI and Thinking
- AI and collaboration
- Similarity and difference between human and artificial forms of creativity and intelligence

Creativity and AI

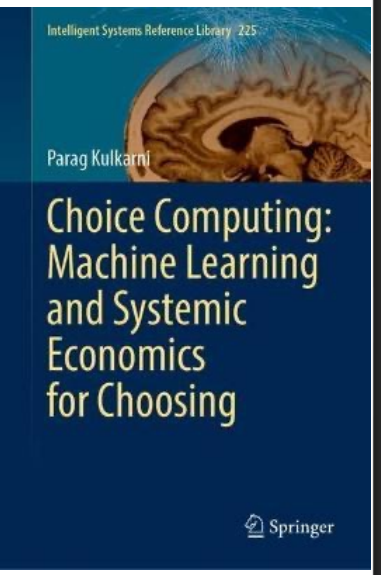
Introduction

- Creativity and Thinking
- Thinking can be taught – Decoding thinking
- Psychological and Historical creativity
- Choice and Creativity
- Creativity is Systemic
- Artificial Creative Intelligence
- Association – Expansion – Summarization and Application

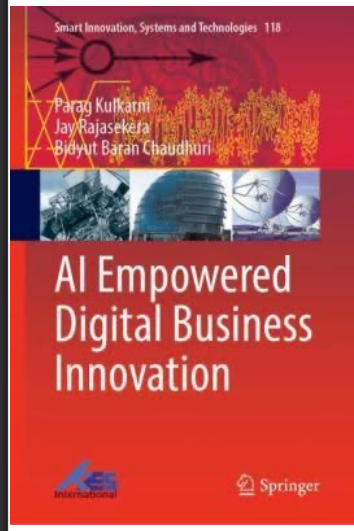
To understand creativity!

- We may need to decode the thought process of creators

Books by Dr. Parag

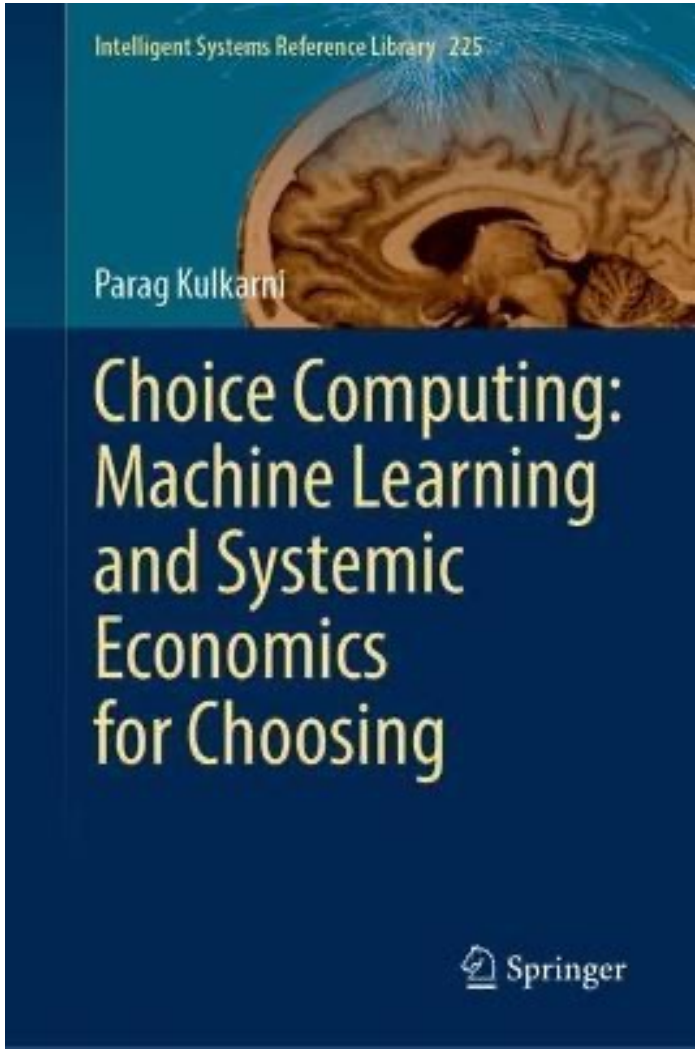






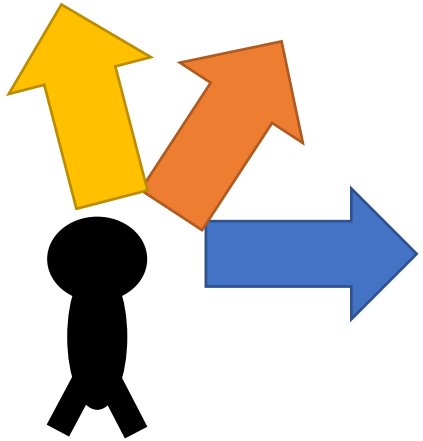
© Parag Kulkarni

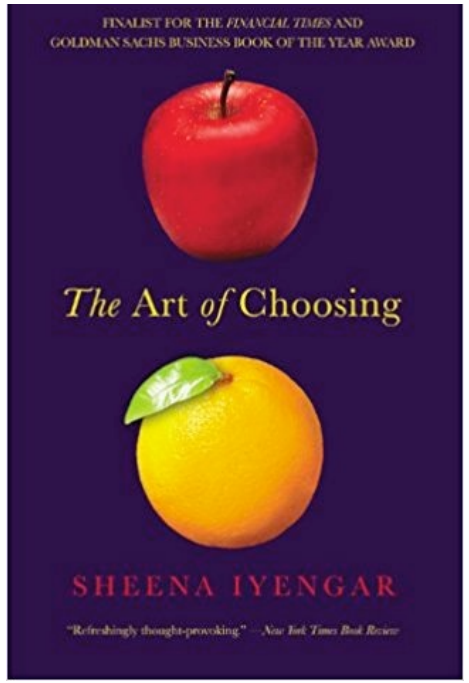
Choice Computing



Choosing and creativity

Choice Engineering



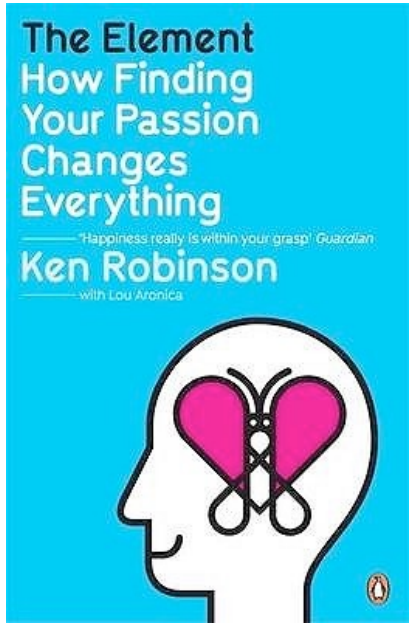


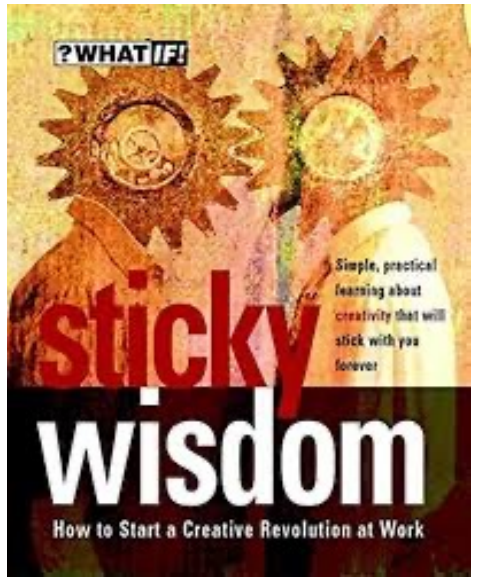
Opportunities…

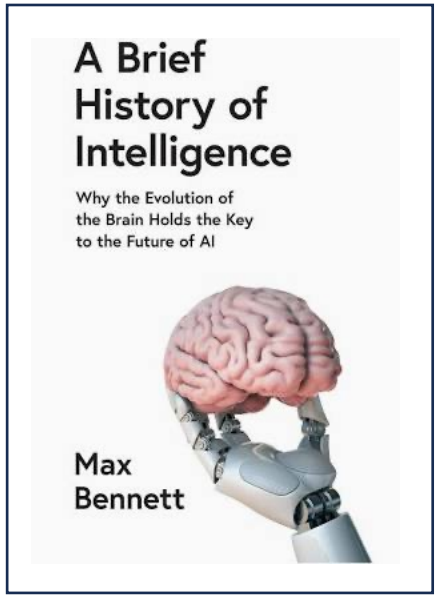
Creative Machine Learning…
(Creativity + Emotions + Consciousness + ?) = Human

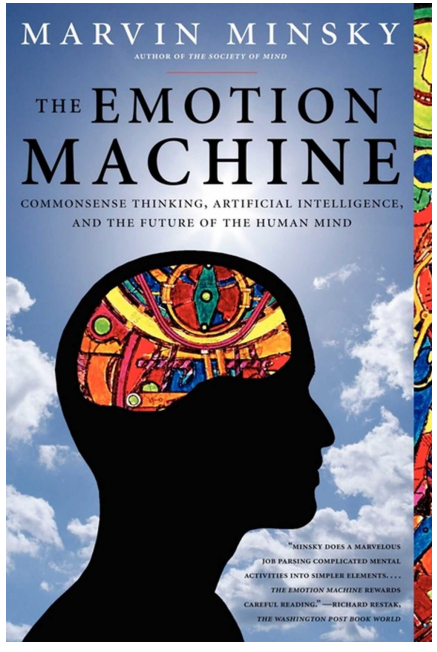
Intelligence is being human?

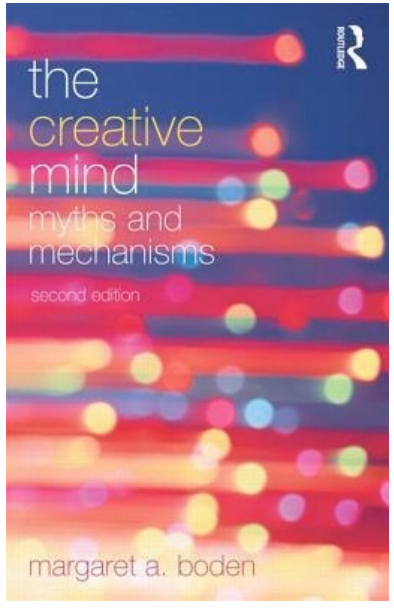
- Keep Learning!!!!!
- Historical creativity and Psychological Creativity
- Evolving systems

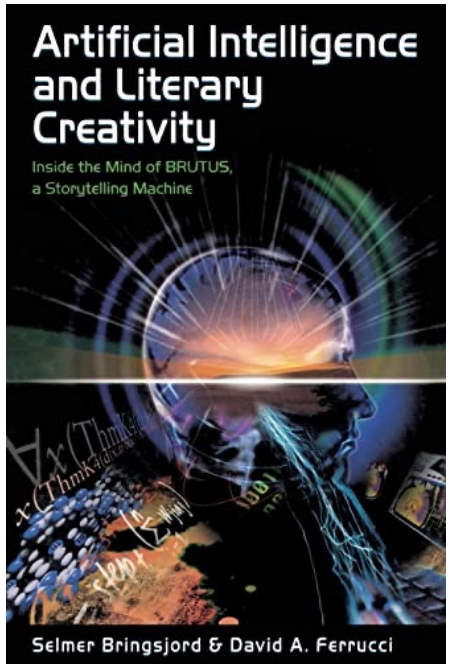


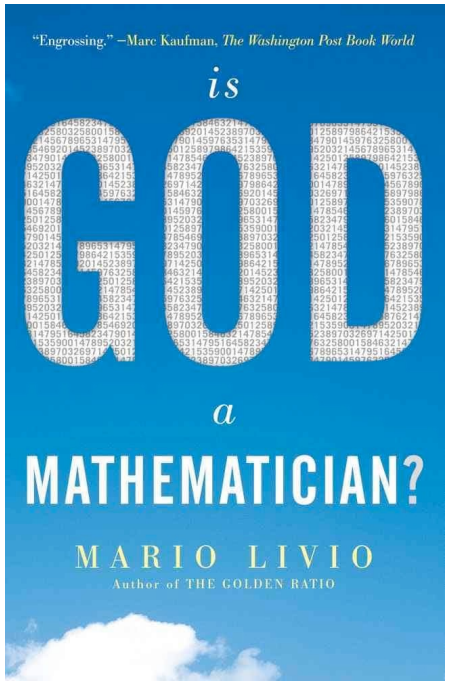




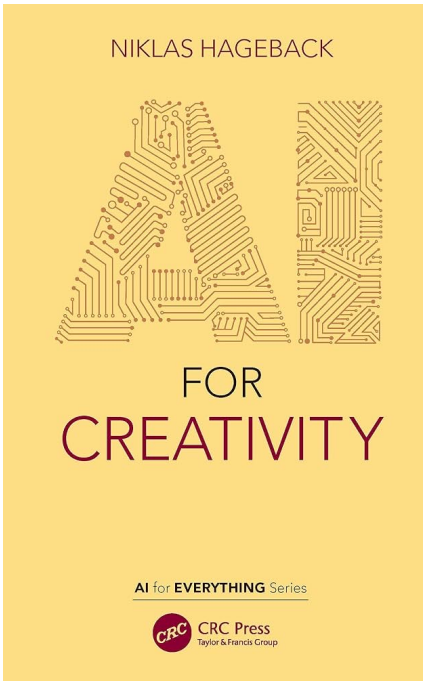


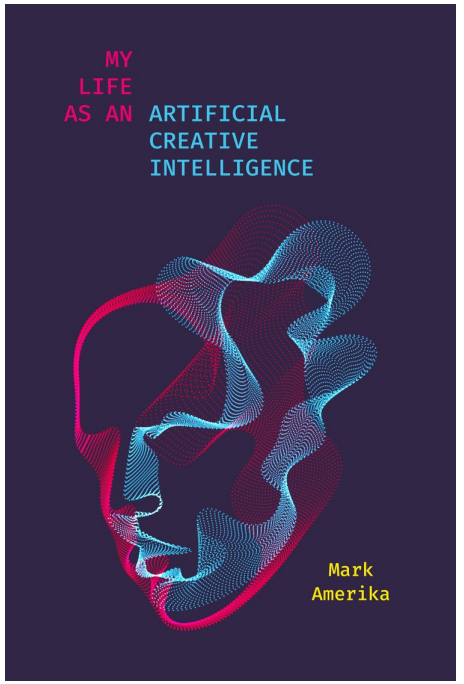












Lateral and creative thinking

Ideas to choice Opportunities...

GENERATION DECODED

How are novel ideas produced at all? Combination, exploration, transformation of conceptual spaces.

MORE THAN SELECTION

Given many candidates, which are worth having — and how do we compute that against a person's actual criteria rather than a generic average?

Choices, not answers

The system reveals a ranked set of genuine options with visible reasoning, rather than issuing one verdict the person can only accept or reject.

Trade-offs made visible

Every selection embeds a weighting of competing goods. Exposing the weighting lets the human disagree with it — which is where their judgement enters.

Human judgement structured, not replaced

The system organises the decision; the person makes it. In creative work this is not a limitation but the correct division of labour.

The Key Concept: Artificial Creative Intelligence

Computational Creativity	The established research field — systems evaluated for creative output against defined criteria
Creative AI	The applied framing — AI as a tool deployed within existing creative industries
Artificial Creative Intelligence	Creativity as a hybrid, distributed process moving across the human–nonhuman spectrum, belonging fully to neither party

counters the traditional creative virtues — authenticity, motivation, empowerment, intentionality — with a different set: intuition, spontaneity, remix. The shift is from creativity as expression of a self to creativity as a process passing through one.

The Key Concept: Lingual spontaneity and the Hybrid Mind (Mark)

*The **surrealists** (**Surrealists** were 20th-century artists and writers who sought to unlock the power of the unconscious mind and bypass rational thoughts) called it psychic automatism: writing without conscious control. Mark Amerika asks whether a generative transformer is doing something structurally similar — and whether AI might develop artificia*

Kerouac modelled his writing after jazz greats like Charlie Parker, treating the typewriter like a saxophone where the writer "blows" unedited thoughts onto the page.

1 The lineage he draws on

Surrealist automatism: · Kerouac's spontaneous prose · jazz improvisation · the Gysin–Burroughs cut-up method. All are techniques for producing text while 'losing consciousness'.

Mind, remixed

Collaboration generates a third authorial presence belonging to neither party. Remixing for the Hybrid Mind.

3 The Hybrid Mind

Not the human author, not the model, but the entity that emerges from their exchange

4 The forward-looking claim

If human intuition is a trained, automatic process, then artificial intuition is not a category error but an engineering horizon. (from Artificial Creative Intelligence Book)

The Key Concept: Terms

Artificial Creative Intelligence (ACI)

Creativity as a hybrid process distributed across human and nonhuman agents

Meta Remix Engine

The artist's own unconscious recombinatory apparatus

Psychic automatism

Surrealist term: composition without conscious control

Cosmotechnical persona

A style-identity understood as technically and cosmologically situated

Onto-operational presence

A mode of being constituted by operation rather than prior substance

Hybrid Mind

The third authorial entity emerging from human–machine exchange

Metamediumystic (Intuitive fluid way to operate)

His coinage for the mystical dimension of working through media

Séance of writing

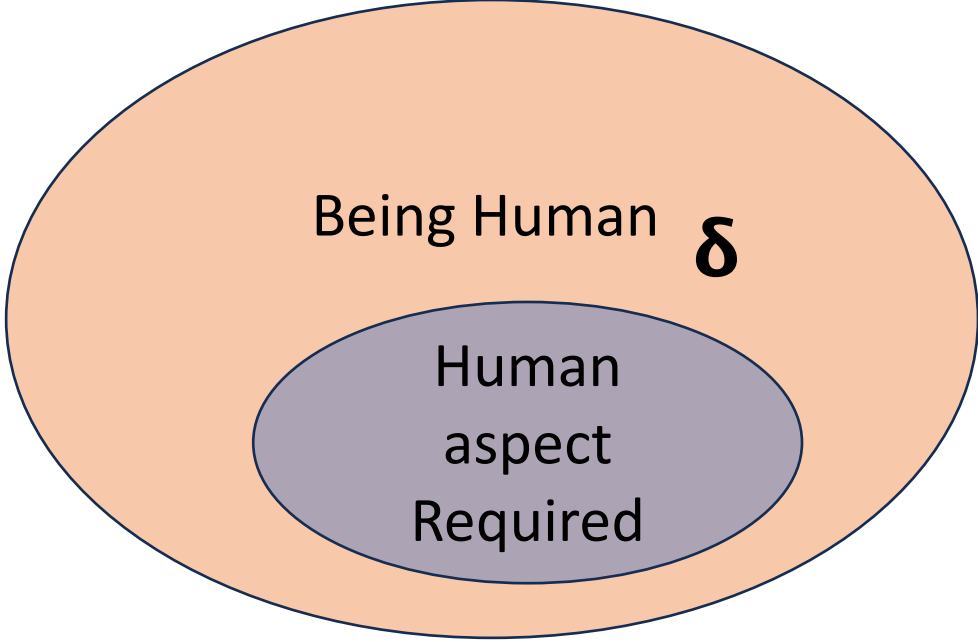
Composition as summoning (Ordering to appear) — mediating presences not one's own

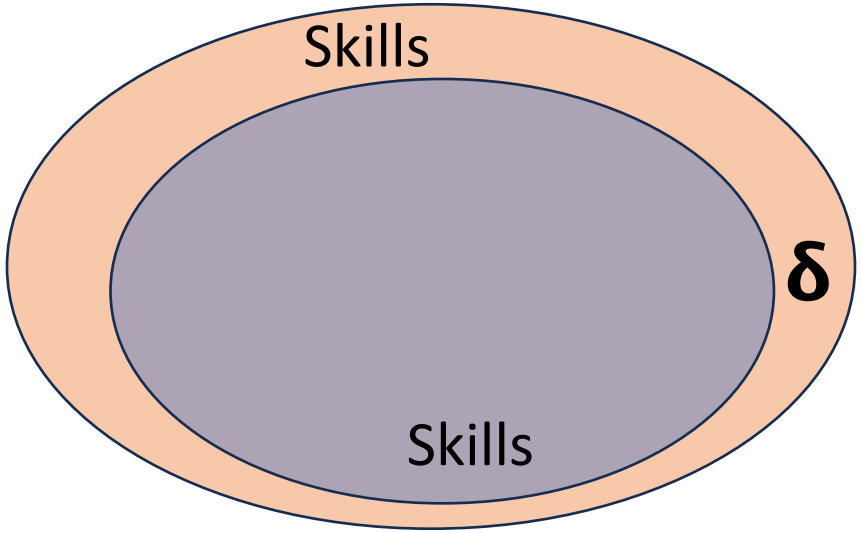
Can/How AI exhibit creativity?

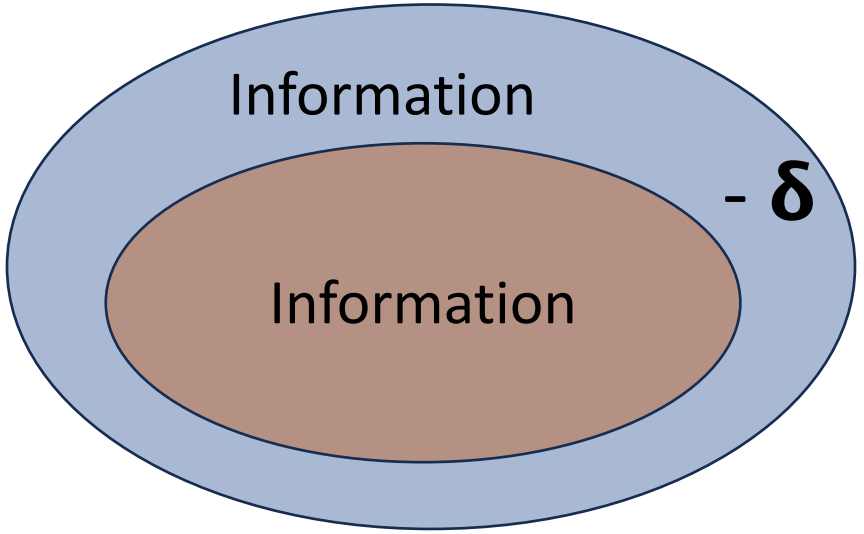
- Search and creativity
- IR and creativity
- Data mining and creativity
- Uncertainty and creativity
- Decision making and creativity

Creativity Gap?

- Being Human (Human Aspects)
- Information processing and accessing
- Skill levels
- It could be positive – it could be negative







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Scenarios

- Your friend draws an interesting new painting!
- Organization launches a product with new features!

Why this subject?

- Thinking outside the box/ Lateral Thinking
- Practicing creativity
- Understanding and applying Critical thinking for and through technology
- Understand AI and IOT in depth
- Economics of AI and IOT
- Applying AI and IOT for business
- Understanding AI
- IOT to improve convenience
- Learning about AI products and making them creative
- Discovering creativity within AI and IOT

Why IOT?

Sensing systems supply what combinational creativity most needs: heterogeneous, continuously refreshed material.

1 New primitives to combine

Combinational creativity needs a rich and varied store. Sensor networks generate exactly that — streams from domains that were never previously juxtaposed.

2 Grounding in the physical

A conceptual space defined over real-world signals is constrained by physics rather than convention, which changes what counts as a valid move within it.

3 Continuous space expansion

New sensor types literally add dimensions to the space. This is the closest practical analogue to transformational creativity available to engineers today.

4 Value becomes measurable

In a physical system, some notions of value — energy saved, comfort delivered, harm avoided — can actually be measured, easing Boden's second requirement.

Cognition: Adding a sensor modality changes the space, not merely the data.

Course Description

Learning is the manifestation of intelligence. While we are trying to make machines intelligent there is a deliberate attempt to embed thinking and creativity into machines. While it is one of the most challenging aspects, it is the most interesting one. This demands understanding fundamentals of creativity and thinking and then finding pathways to embed it into machines.

This course introduces the concepts related to human thinking and deliberates on concepts like meta reasoning and meta-thinking to find out how creativity can be embedded into machines. Most importantly the course tries to throw light on how concepts of creativity and human thinking can be used in a more objective and humanized way for AI innovations and AI-Human co-working for creative activities.

Objectives:

1. Learn fundamentals of creativity
2. Understand creativity opportunities while building AI and ML based products and solutions
3. Understand thinking patterns
4. Learn about psychology and AI
5. Learning different concepts about thinking and associating it to solve problems using AI and IOT
6. Develop concepts related to creativity based ethical compass for AI innovations

Syllabus

Class	Contents
1	Thinking can be taught? Decoding thinking from computational perspective · Human thinking and creativity
2	Principles and practices of creativity (different verticals) · Human creativity examples, machine creativity examples
3	Types of creativity: Psychological and Historical Creativity · Embedding creativity in machines · Map of Mind, mind maps and association
4	Creative thinking with data mining · Enhancing thought process (storytelling, innovation heuristics, behaviour models) · Empowering AI innovations with human creativity concepts
5	Selection and Choosing: Curation in a world of excess · Creativity, selection and classification · Choosing paradigms, classification and clustering
6	Basic machine learning and human learning paradigms · Lateral thinking: can machines think differently? · Thinking and machine learning
7	Make data speak intelligently and creatively · Data, patterns and creativity pointers
8	Approaching innovations in a data mining way · Entropy, uncertainty and creativity
9	Creative Connections: the art of creative inferencing · Connectionist learning (neural approaches)
10	Simple agents and intelligent agents · From agents to creative agents · Innovations: making ideas work using AI and ML
11	Planning and creativity · Pathways to systemic innovations: a holistic approach · Study of research papers on AI in creativity domains · Review and Quiz
12	Enhancing machine learning paradigms with human learning paradigms and creativity · Embedding meta-thinking approaches
13	AI–IoT innovations: conceptual space expansion approach · Understanding AI applications and creative solutions

Syllabus

Class	Contents
14	Data-driven approaches: association, differentiation and expansion · Statistical techniques and creativity
15	Learning-centred approaches: evolution, exploration and generation · Meta-learning in humans and machines
16	Bio-inspired approaches: enhancing neural and group approaches
17	Data capture: cognitive aspects of acquisition and visualization · Computational creativity in action: inventions to innovation · Modeling creativity and cognition
18	Designing systems and approaching AI innovation · Intelligent Agents: sense, learn, infer, act — Part One (Consciousness?) · Agentic Systems — Part Two (Consciousness?)
19	Holistic and systemic paradigm · Hybrid approaches, systemic interdependence, system and conceptual space boundaries · Identifying regions for expansion
20	AI tools for creativity · AI product case studies
21	Genetic, generative and exploratory learning paradigms for creativity · AI and ML opportunities in creativity space (Review and Quiz) · Creativity pointers for social computing
22	IoT as more than data collection: making devices smart · Thought translation devices · AI helping humans in creative tasks (role of IoT)
23	Thought experiments, designing simple systems, group exercises · AI co-working with humans on creative projects
24	Simple AI + IoT to build creative solutions · Developing applications with creative agents
25	Ethical aspects when AI and humans come together in creative disciplines
26	Miscellaneous concepts and looking ahead · Connectionism, cognitive flexibility, generalisation, humanized AI, future of AI · Learning across domains to deliver outcomes

Creative computing

- How machines can help humans to exhibit creative behavior?
- How machines can learn from human creativity to exhibit human like creative behavior?
- How machines and humans could co-create the creative actions/products or artefacts.

Computational Creativity

- *Creativity is a process involving the generation of new ideas or concepts, or new associations between existing ideas or concepts*

  -- Wikipedia.

- Humans are creative. Ability to think out of the box.
- Goal of Computational Creativity: To model, simulate or replicate creativity using a computer.

Why Creativity?

- To construct a program/computer capable of human-level creativity.
- To better understand human creativity and to formulate an algorithmic perspective on creative behavior in humans.
- To design programs that can enhance human creativity without necessarily being creative themselves.

Critical Vs Creative Thinking

Critical thinkers:

- Care that their beliefs be true and that their decisions be justified; that is, care to "get it right“ to the extent possible. This includes the dispositions to

  - ✓ **Seek alternative hypotheses, explanations, conclusions, plans, sources, etc.,and be open to them**
  - ✓ **Endorse a position to the extent that, but only to the extent that, it is justified by the information that is available**
  - ✓ **Be well informed**
  - ✓ **Consider seriously points of view other than their own**

Characteristics II

Critical thinking:

- Clarity and Honesty

✓**Clarity and Precision**
✓**Focus on, the conclusion or question**
✓**Overall considerations of opinions**
✓**Systemic components**
✓**Awareness of basic beliefs**
✓**Educating other agents**

Exercise

Choose a creative artefact you admire — a piece of music, a product, a proof, a building. Classify the creativity involved using Boden's three forms, and defend the classification.

1 Name the conceptual space

What were the rules, conventions and constraints in force before this artefact existed? Be specific — vagueness here makes the rest impossible.

2 Classify the move

Was it a combination, an exploration, or a transformation? Most interesting cases are contested — argue for one and note what would count against you.

3 Locate the value judgement

Who decided it was good, on what basis, and when? Was the value recognised immediately or only later?

4 Ask the machine question

Which part could a system have done? Which part could it not? Be precise about why — this is the exercise's actual purpose.

1
Begin with a Problem...

2
Identify Learning Opportunities...

3
ML Model is the key...

4
Model to value Creation...

Enhance The Model...

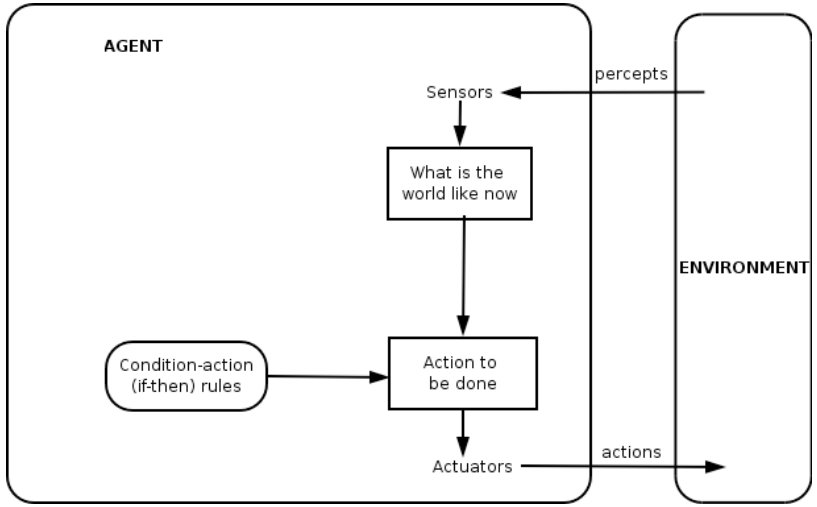
5
Generalization and Continuous Learning...

6
Generative, Deep and Exploratory...



Agent

- Perceiving its environment through sensors and action on environment through actuators
- Percept sequence
- Agent Function
- Rationality – Rational agent

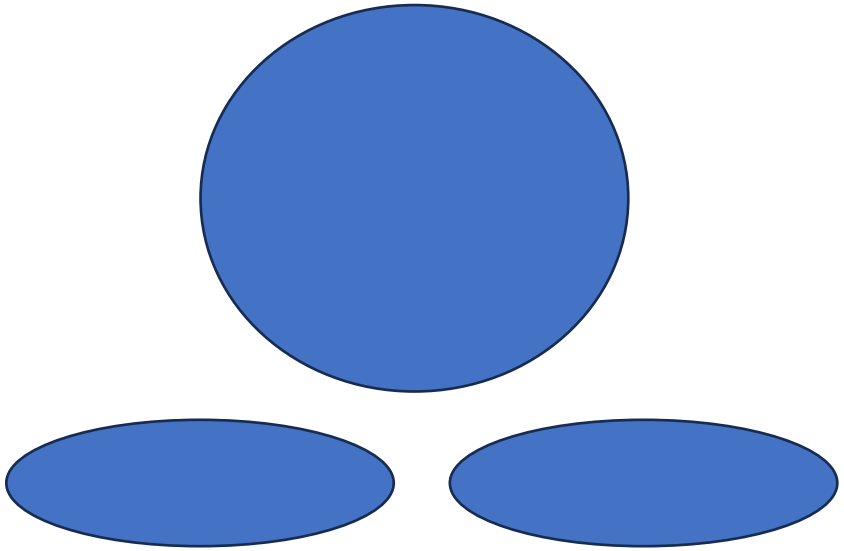


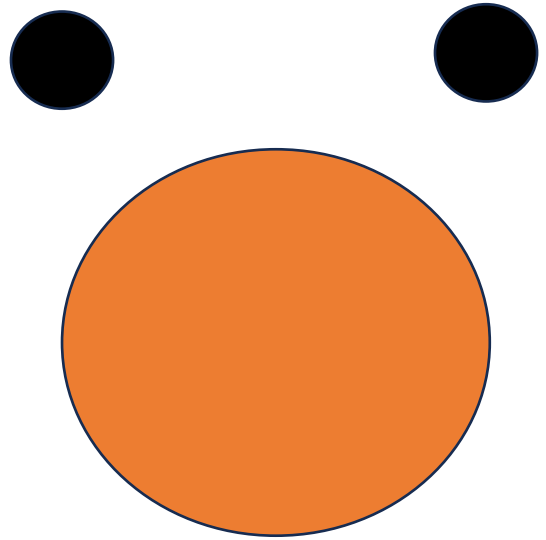
PEAS

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- For creative agent what could be PEAS
  - +Performance measure
  - +Environment
  - +Actuators
  - +Sensors



Creativity





- Creative Agents Creative agents are poised to revolutionize content creation by generating and editing images, videos, audio, and copy while adhering to brand guidelines. Google introduced AI-powered Workspace apps, including auto-generated videos, as part of this initiative. With creative agents, businesses of all sizes will have access to advanced content creation capabilities, democratizing the creative process.

- Repurpose
- Complete differently
- Story formation
- Associate words, optimal word, repurpose words
- Building blocks to design different shapes
- Incremental design – evolutionary design
- Exploratory design
- Substitute, eliminate, what is minimal, what is optimal, adaptation, combine, expand, Reverse
- Associate, analogy

TURING TEST



Machine Response



Human Responder



Interrogator / investigator

CREATIVITY TEST



Machine Response



Human Responder



Interrogator / investigator

Creativity evaluator

Some responses may be creative!

Cannot detect: Who is the originator of creative response while some of them are also coming from machine

Different approaches and views of creativity:

- Nomothetic versus idiographic view: The nomothetic approach focuses on universals of creativity while the idiographic focuses on individual differences and uniqueness of creative individuals
- Mental versus social phenomenon: Creativity is seen as a mental or psychological phenomenon for the generation of new ideas or as the result of an audience's appreciation of a novel idea.
- Improbabilist versus impossibilist approach: The improbabilist approaches see creativity as a novel combination of familiar ideas while the impossibilists consider creativity as new ideas never appeared before. This is possible thanks to a transformation of the conceptual space.
- Rational versus irrational process: The rational approaches include logic and evolution. Logic sees creativity in processes like abduction (guessing a conclusion given some premises), eduction – inferring (relate properties of different objects), deduction and induction (relate properties from general to specific and vice versa). Evolutionary approaches instead see creativity as an adaptation to the environment motivated by problem solving while the irrational approach focuses on the emotional part of creativity.

General Introduction

- Creative thinking
- AI and IOT innovation
- What is significance of this subject?

Essential Characteristics of a Creative Idea

Ideas of Newell, Shaw and Simon

- Novelty and usefulness (either for the individual or society)
- Rejection of previous ideas
- Results from intense motivation and persistence
- Clarification of vague ideas

Margaret Boden's view

- P-Creativity (Psychological or Individual)
- H-Creativity (Historical or Collective)

How Generative AI Can Augment Human Creativity

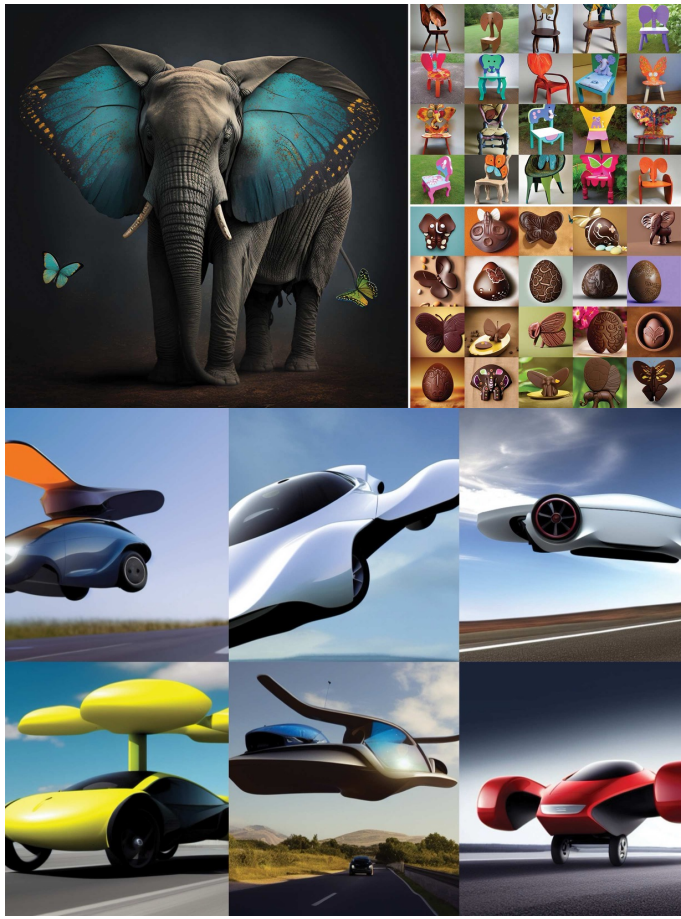
- Companies have used crowdsourcing and idea competitions to involve outsiders in the innovation process
- Crowdsourcing, for instance, may produce a flood of ideas, many of which end up being dumped or disregarded because companies have no efficient way to evaluate them or merge incomplete or minor ideas that could prove potent in combination.



https://hbr.org/2023/07/how-generative-ai-can-augment-human-creativity

Why AI in creativity?

- Boundary (flexible) Promote Divergent Thinking
- One idea + another: Concept + Theme
- Bias is too biased! Challenge Expertise Bias
- Idea evaluation bottleneck! - Assist in Idea Evaluation
- Improvement and refinement issues



But!

- “It’s easy for AI to come up with something novel just randomly. But it’s very hard to come up with something that is novel and unexpected and useful.”

— John Smith, Manager of Multimedia and Vision at IBM Research

- Can we take what humans think is beautiful and creative and try to put that into an algorithm? I don't think it's going to be possible for quite a while.”

— Jason Toy, CEO, Somatic

- What I'm looking for out of cognitive systems is not just another form of computing but something that actually creates a presence in our life and through that presence is able to inspire us.”

— Rob High, Vice President and CTO, IBM Watson

Research Papers

- https://www.sciencedirect.com/science/article/pii/S0004370298000551
- https://onlinelibrary.wiley.com/doi/full/10.1002/jocb.597
- https://www.researchgate.net/publication/371956576_Using_Artificial_Intelligence_for_enhancing_Human_Creativity/link/649e7ec8b9ed6874a5e9a1f6/download
- https://www.degruyter.com/document/doi/10.1515/9783110198560/html?lang=en
- https://iopscience.iop.org/article/10.1088/1742-6596/1760/1/012032/meta

Creativity

- AI alone
- AI + AI
- AI + Human
- Multi machine – multi-thinkers

Creativity or illusion of creativity?

- Do we need to redefine creativity?
- Do we need tool to differentiate human and machine creativity?

Business – creativity and Machines

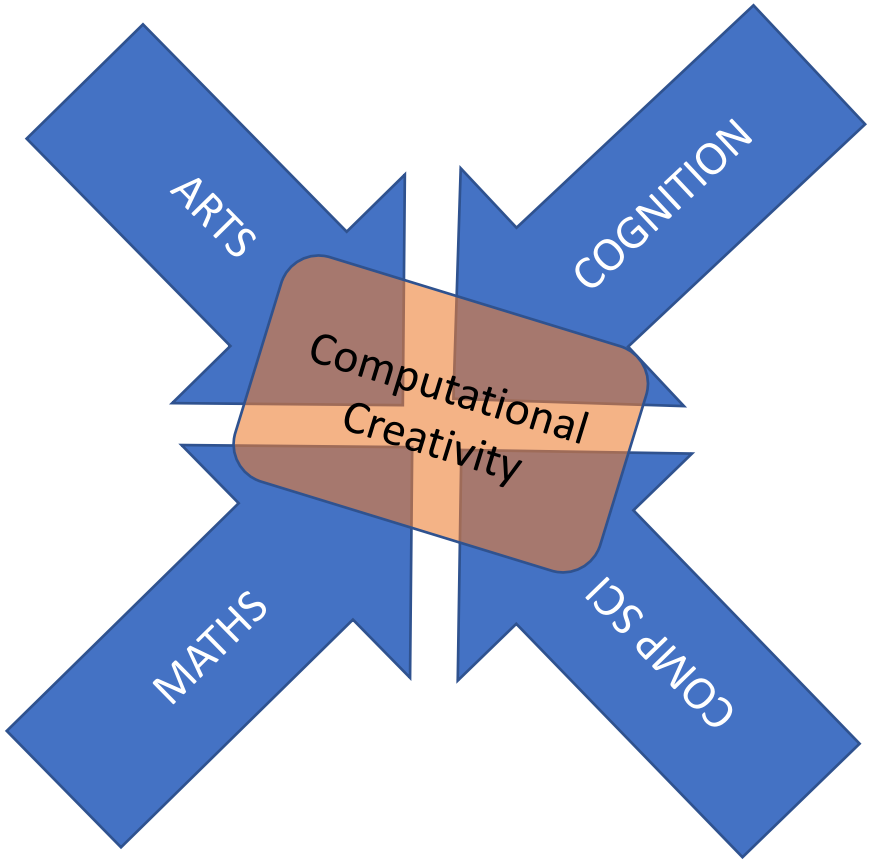
- Synthetic creativity on demand, as currently generated by AI, is certainly a boon to business and marketing. Recent examples include:
- AI-enhanced advertising: Ogilvy Paris used Dall-E to create an AI version of Vermeer's *The Milkmaid* for Nestle yoghurts.
- AI-designed furniture: Kartell, Philippe Starck and Autodesk collaborated with AI to create the first chair designed using AI for sustainable manufacturing.
- AI-augmented fashion styling: Stitch Fix utilised AI to capture personalised visualisations of clothing based on requested customer preferences such as colour, fabric and style.

https://www.weforum.org/agenda/2023/06/ai-cannot-replace-human-creativity/

Quick assignment

- Words those are indicative of creativity

Can you represent creativity mathematically?



Combination of:

- Computational Systems and
- Human creator

Autonomous Vs semi-autonomous

- Is it strong vs weak?

IDEA Canvas-1: Tell one instance of creativity?



Is creativity?

- Exploration?
- Evolution?
- Change?
- Option Creation?
- Association?
- Combination?
- Knowing ahead of time?
- Surprise / Uncertainty / Anti-Pattern?

Formalizing the Philosophical Concepts

Exploration: Within a conceptual space

Transformation: Out of the box(space)

Formalization :

- Conceptual Space C a strict subset of set of all concepts U
- Axiom1:Every concept c is a distinct member of U
- Axiom2:Every conceptual space includes F (empty concept)

Formalization (contd…)

- Rules : R(Existence in a space) and T(Transformation in a space) R , T subsets of L
- Interpretation function: [.] and Traversal function: <.>
- Exploratory conceptual space as a tuple : (U,L,R,T,E)
- Beginning of exploratory creative process : <R + T>({F})
- Evaluating concepts (E) : E(C) = value of the conceptual space.

Where is the AI?

- (RACTER, 1984, https://www.poem-generator.org.uk) generates poems, stories, etc.
  - Rules to be broken or adhered
  - Syntax directives
  - Sentences too bizarre at first look
  - Deeper meaning on repeated thought
- Creativity is in the reader's mind
- Sentences become insignificant soon
- Need for more control

Auto-poem generation

- **Template Based Poetry Generation:** Templates of poetry forms are filled with words that suit the defined constraints (either syntactic, rhythmic or both). After using the Poetry Creator and looking at several generated results, it is clear that the system follows this approach.
- **Generate and Test Approaches**: Random word sequences are produced according to formal requirements, that may involve metric, other formal constraints and semantic constraints. Manurung's chart system, the WASP system and the generate and test strategy of Tra-La-Lyrics follow this approach.
- **Evolutionary Approaches**: Poetry generation is based on evolutionary computation. POEVOLVE and Manu- rung's McGonnagall are two systems that follow an evolutionary approach.
- **Case-Based Reasoning Approaches**: Existing poems are retrieved, considering a target message provided by the user, and are then adapted to fit the required con- tent. ASPERA and COLIBRI are examples of this approach

Ref: Gonçalo Oliveira, Hugo. (2009). Automatic generation of poetry: an overview.

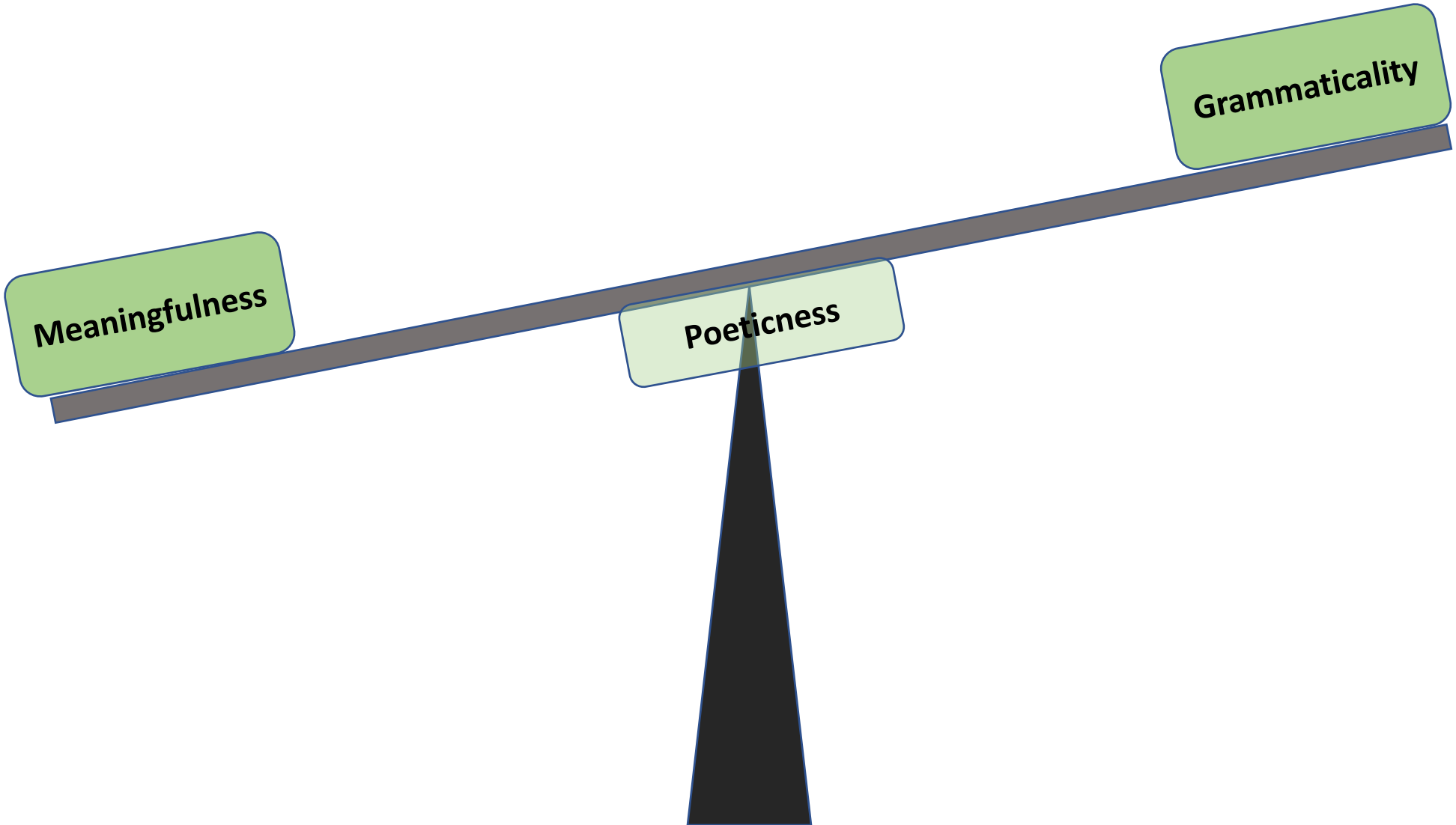
Need for Flexibility

- Rule-based systems are monotonous
- Example: Generating a story (TALE-SPIN)
  - Each object tries to satisfy its goals
  - Creativity is shown only when the plot turns an unexpected way
  - Object need not try to reach goals at every step
- Solution: Use the “meta-” approach
- Develop rules for rules

Rhyme – meaning and art in business

- **Meaningfulness**: convey a conceptual message, which is meaningful under some interpretation.
- **Grammaticality**: obey linguistic conventions prescribed by a given grammar and lexicon.
- **Poeticness**: exhibit poetic features.

Striking the creativity balance!

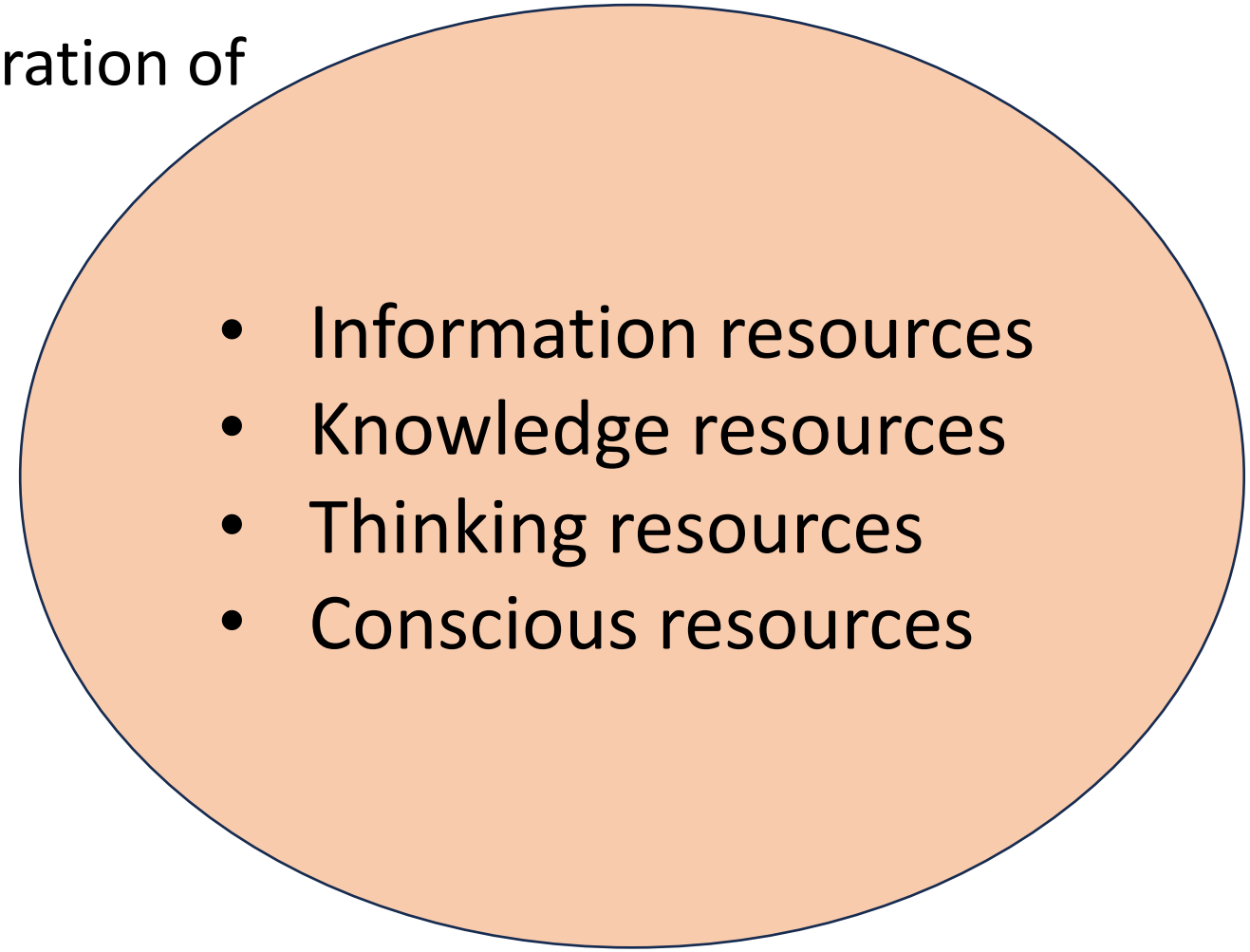


Examples

- The ALAMO group
- Manurung’s chart system
- POEVOLVE
- WASP
- ASPERA
- McGonnagall

Thinking about emotions

- Collaboration of



What could be creative alterations here?

GRASS CUTTER



- Shyama developed a grass cutter
- Shyama on tricycles

Creativity is the thought process!
Approved by ?

Quick assignment

- Scenario where you thought machine is creative!

Creativity Result from Interactions

- System
- Assessment
- Validation

Creativity

- A process by which a symbolic domain in culture is changed

Arrange jumbled words

I think do can taught it is simple and thinking be

Possible combinations and which one is creative?

Thinks about the words

What Can be intelligent? Can they novel and endorse creativity of individuals?

- Products
- Services
- Actions
- Interactions
- Expressions
- Teams

Quick assignment

- How could you formulate creativity test for humans?
- How can you get rid of subjectivity in creativity? Is I possible?

TURING TEST



Machine Response



Human Responder



Interrogator / investigator

CREATIVITY TEST



Machine Response



Human Responder



Interrogator / investigator

Creativity evaluator

Some responses may be creative!

Cannot detect: Who is the originator of creative response while some of them are also coming from machine

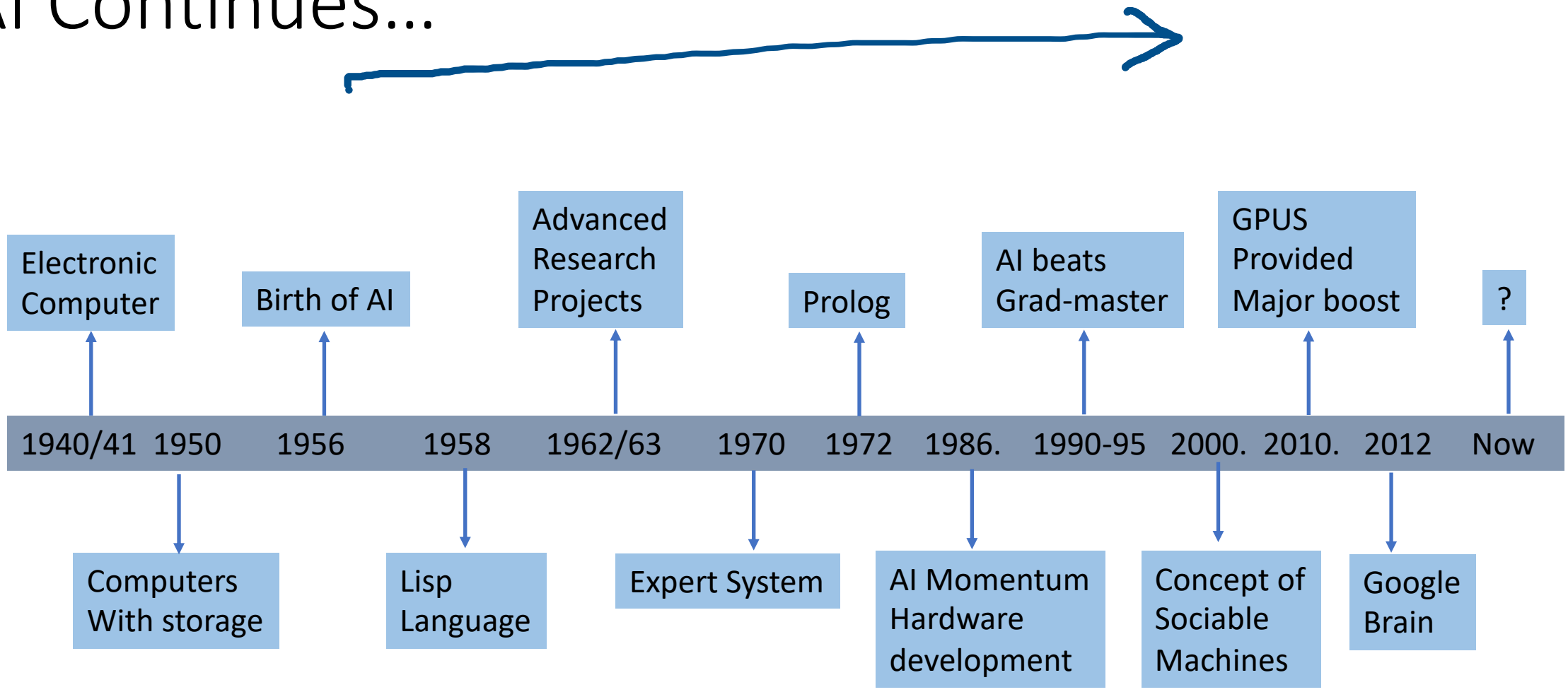
Different approaches and views of creativity:

- Nomothetic versus idiographic view: The nomothetic approach focuses on universals of creativity while the idiographic focuses on individual differences and uniqueness of creative individuals
- Mental versus social phenomenon: Creativity is seen as a mental or psychological phenomenon for the generation of new ideas or as the result of an audience's appreciation of a novel idea.
- Improbabilist versus impossibilist approach: The improbabilist approaches see creativity as a novel combination of familiar ideas while the impossibilists consider creativity as new ideas never appeared before. This is possible thanks to a transformation of the conceptual space.
- Rational versus irrational process: The rational approaches include logic and evolution. Logic sees creativity in processes like abduction (guessing a conclusion given some premises), eduction – inferring (relate properties of different objects), deduction and induction (relate properties from general to specific and vice versa). Evolutionary approaches instead see creativity as an adaptation to the environment motivated by problem solving while the irrational approach focuses on the emotional part of creativity.

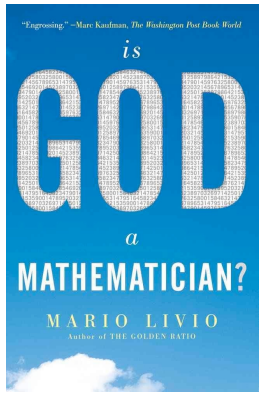
What can be more intelligent?



AI Continues…



Thinking…


"Engrossing." —Marc Kaufman, The Washington Post Book World
is
GOD
a
MATHEMATICIAN?
MARIO LIVIO
Author of THE GOLDEN RATIO


EDWARD
DE BONO
TEACHING THINKING

What is thinking?

- Is it about reasoning?
- Is it about finding relationship?
- Is it about finding process to solve problem
- Thinking is organizing and reorganizing known facts and learning and extrapolating them in the context of problem

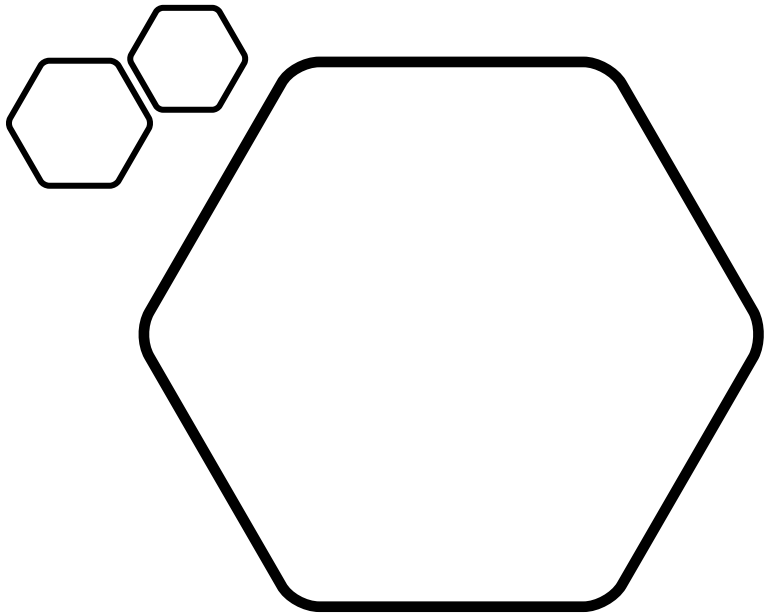
Types of Thinking

- Perpetual or concrete
- Conceptual or abstract
- Logical with reasoning
- Creative
- Problem based – problem solving oriented
- Convergent Vs Divergent

Convergent Vs Divergent thinking

- **Convergent Thinking** focuses on attaining solution of the problem – it aims to converge to solution and hence many times constrained
- In contrast, **divergent thinking** is about looking in different directions – opening mind and going beyond boundaries – it also about trying out multiple solutions for given problem

Thought Experiments – Creative Learning with limited Success

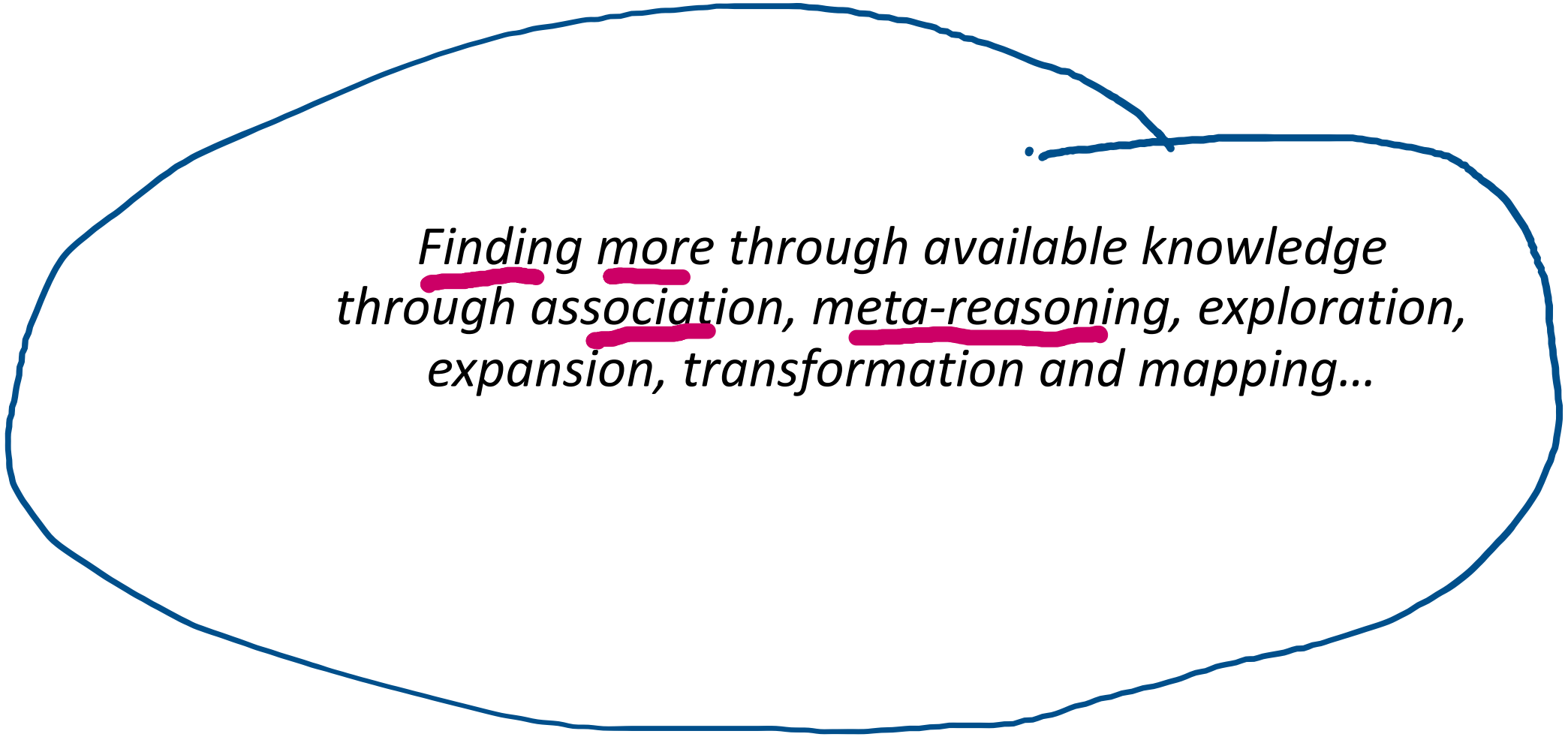


- Taking contrary view
- Thinking one subject in terms of another subject
- Taking multiple perspectives
- Hunting opportunities to think differently
- Resulting new relevant(?) combinations using existing ones

Learning is building ability to produce the expected results

IDEA Canvas-2: How would you define learning? When do you say you learnt something?




Finding more through available knowledge through association, meta-reasoning, exploration, expansion, transformation and mapping…

Thinking from Computing Perspective

- Machine Thinking?
- Cognition, thinking and Computing

Summary

- Objectives
- Thinking
- Computing

Thanks